

EXPT NO:7	OVER-PLOTTING REDUCTION TECHNIQUES
DATE: 09.02.2026	

PRE-LAB QUESTIONS

1. **Why is over-plotting common in big data visualization?**
Because millions of data points are plotted in limited screen space, causing points to overlap and hide patterns.
2. **How does data density affect perception?**
High density creates dark clusters that dominate attention, while sparse areas may be overlooked, leading to biased interpretation.
3. **What trade-offs exist between detail and clarity?**
Showing all data preserves detail but reduces readability; summarizing data improves clarity but may hide rare or important patterns.
4. **How do AI datasets increase visualization complexity?**
AI datasets are large, high-dimensional, and continuous, making direct plotting cluttered and harder to interpret.
5. **Why is over-plotting a serious analytical risk?**
It can conceal trends, outliers, and relationships, leading to incorrect conclusions and poor decision-making.

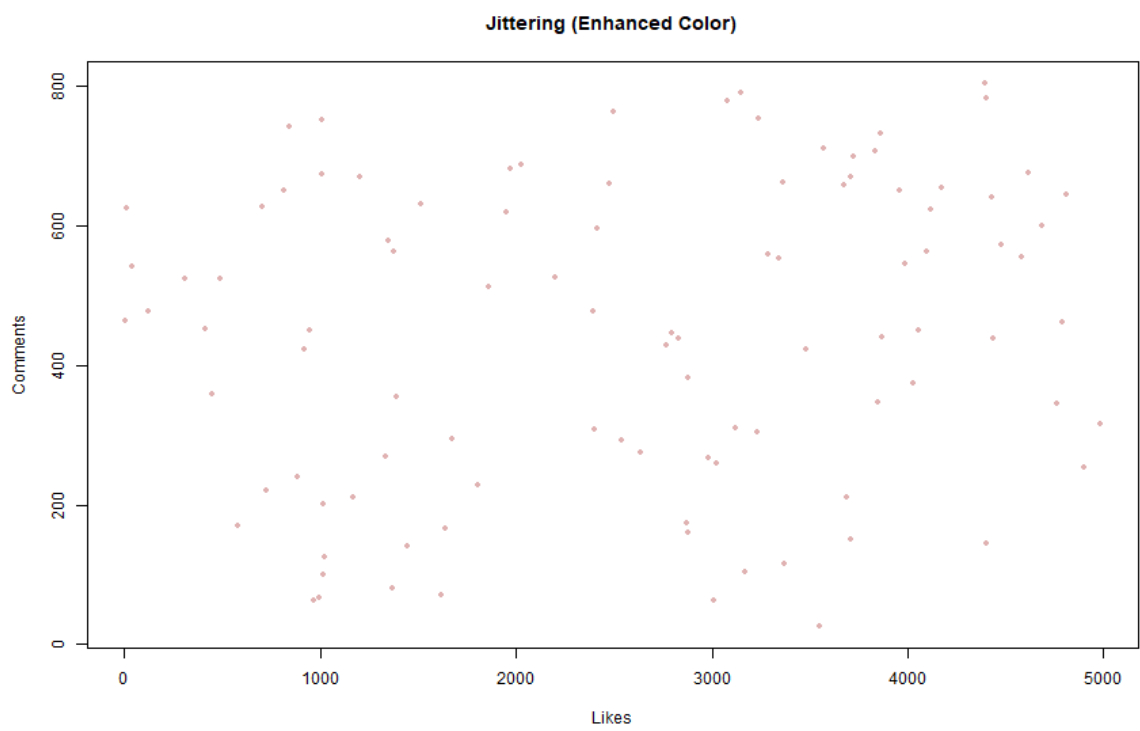
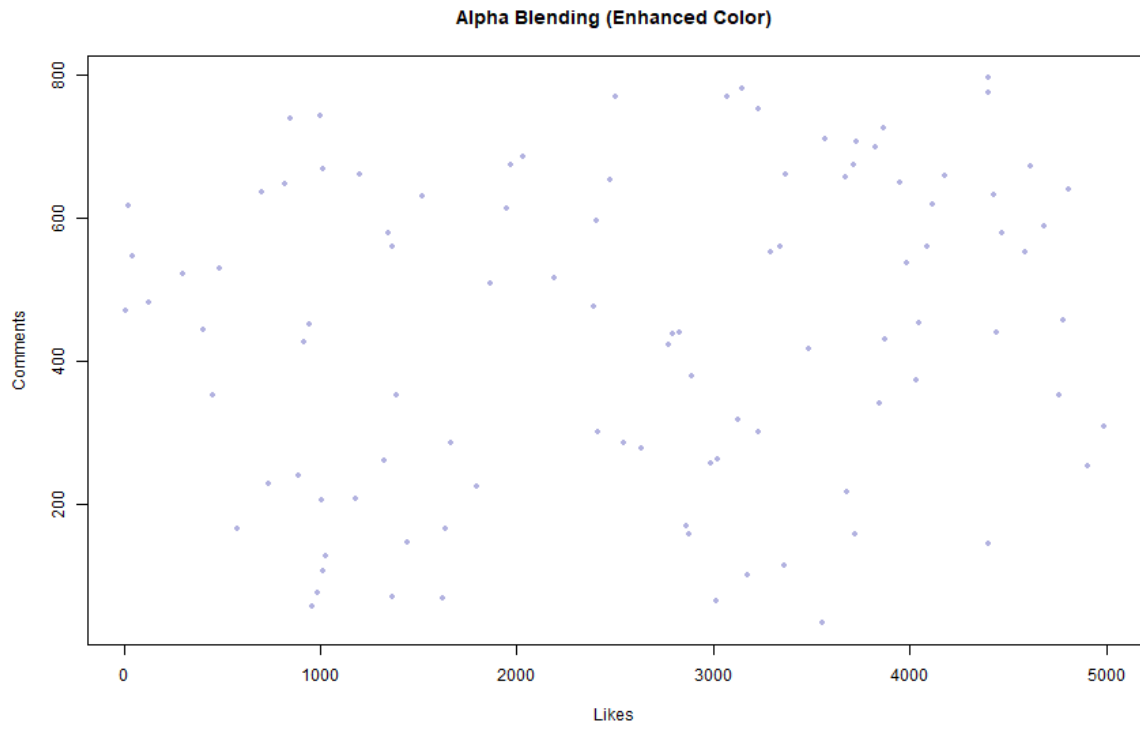
OBJECTIVE : To apply techniques that reduce visual clutter in large-scale datasets.

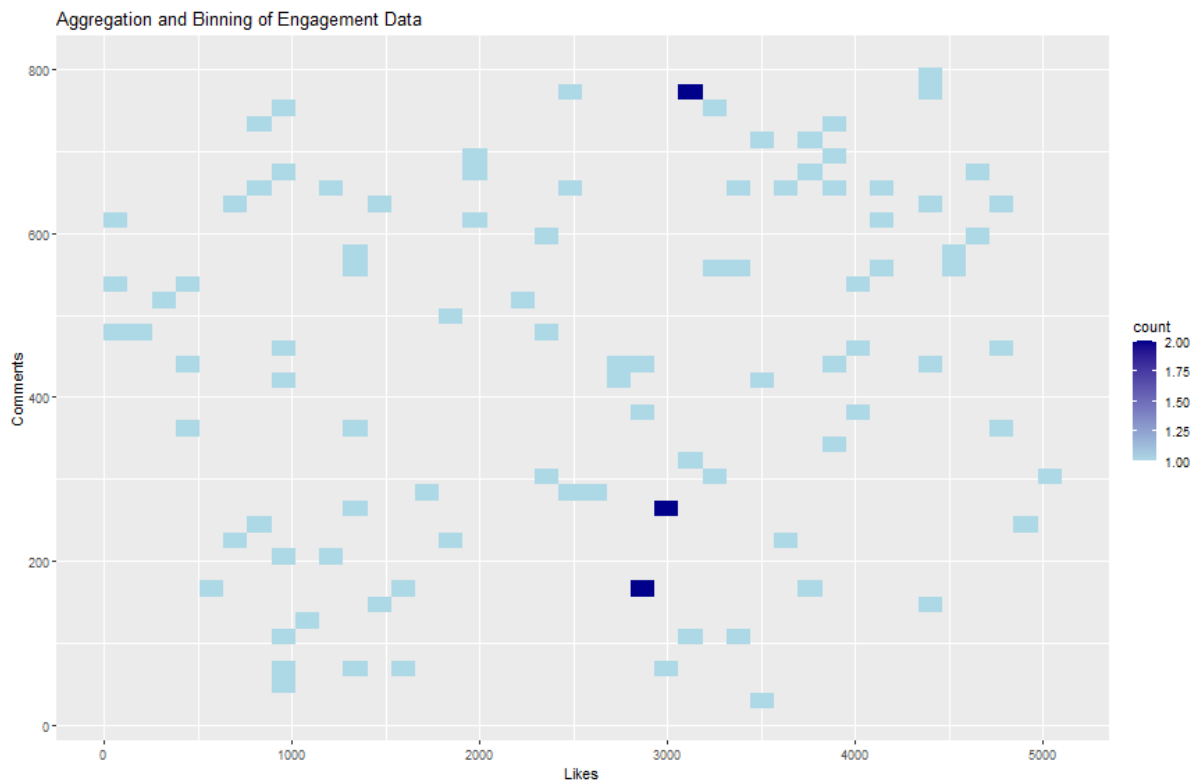
SCENARIO A social media analytics company visualizes millions of user interactions to study engagement patterns.

IN-LAB TASKS (Using R Language) • Apply alpha blending • Implement jittering techniques • Use aggregation and binning

CODE:

```
1 # =====
2 # Roll No: 23BAD069
3 # Experiment: Reducing Visual Clutter
4 # =====
5 # Load libraries
6 library(readxl)
7 library(ggplot2)
8 # =====
9 # 1. Import Dataset
10 # =====
11 df <- read_excel("C:/Users/student/Downloads/7.social_media_interactions.xlsx")
12 # =====
13 # 2. Basic Cleaning
14 # =====
15 df <- na.omit(df)
16 df <- unique(df)
17 # =====
18 # 3. Select Engagement Columns
19 # =====
20 x <- df$Likes
21 y <- df$Comments
22 # =====
23 # 4. Alpha Blending
24 # =====
25 plot(df$Likes, df$Comments,
26      pch = 16,
27      cex = 0.8,
28      col = rgb(0, 0, 0.6, alpha = 0.3), # darker blue
29      xlab = "Likes",
30      ylab = "Comments",
31      main = "Alpha Blending (Enhanced Color)")
32 # =====
33 # 5. Jittering
34 # =====
35 plot(jitter(df$Likes, amount = 10),
36      jitter(df$Comments, amount = 10),
37      pch = 16,
38      cex = 0.8,
39      col = rgb(0.6, 0, 0, alpha = 0.3), # darker red
40      xlab = "Likes",
41      ylab = "Comments",
42      main = "Jittering (Enhanced Color)")
43 # =====
44 # 6. Aggregation & Binning
45 # =====
46 ggplot(df, aes(Likes, Comments)) +
47   stat_bin2d(bins = 40) +
48   scale_fill_gradient(low = "lightblue", high = "darkblue") +
49   labs(title = "Aggregation and Binning of Engagement Data",
50        x = "Likes",
51        y = "Comments")
52
```





POST-LAB QUESTIONS

1. **Which technique provided the best clarity and why?**

Aggregation and binning gave the best clarity because they summarize millions of points into density regions, revealing overall engagement patterns clearly.

2. **How does over-plotting distort analytical conclusions?**

It hides true data distribution, masks clusters and outliers, and can falsely suggest uniform or misleading trends.

3. **When should aggregation be preferred over raw plotting?**

When datasets are very large or dense and individual points overlap excessively, making raw plots unreadable.

4. **How do these techniques support scalable AI analytics?**

They enable efficient visualization of massive AI datasets by preserving patterns while reducing noise and computational load.

5. **Explain real-world consequences of ignoring over-plotting.**

Misinterpreting user behavior, incorrect model insights, poor business decisions, and flawed AI-driven recommendations can result.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		